

From Lists to Oracles: Competition for Generative Search Visibility and Long-Tail Entry

Xing Hu

Faculty of Business and Economics, The University of Hong Kong

xinghu@hku.hk

Abstract. The transition from ranked-list search engines to generative AI “oracles” represents a fundamental shift in the allocation mechanism of the digital attention economy. We develop a game-theoretic framework to analyze how the move from divisible visibility (lists) to winner-takes-all exclusion (answers) reshapes competitive incentives and market structure. We model the competition between a high-authority Head firm and a low-authority Tail firm, where the platform’s scoring rule aggregates prior authority and query-specific evidence via a strategic authority weight, while firms exert costly optimization effort (GEO) to signal their intrinsic relevance. Our results demonstrate that Generative Search drives a surge-or-retreat equilibrium leading to a bifurcated market: in broad queries with commoditized relevance, the game intensifies Matthew-Effect concentration; in niche queries, sufficiently specialized SMEs can overturn incumbency to capture the market via a long-tail flip. Furthermore, we uncover a dual-endowment reversal: intrinsic relevance acts as a substitute for effort once a specialist secures the answer slot, whereas an authority disadvantage forces the SME to escalate effort to deter entry. Finally, for platforms and policymakers, we identify the authority weight as a critical market-design lever that dictates competitiveness, redistributes visibility, and determines the quality of information delivered to the user.

Key words: Generative AI Search; Generative Engine Optimization (GEO); Answer-Level Visibility; Visibility Inequality; Market Structure; Search Engine Optimization (SEO), Long Tail, Mathew Effect

1. Introduction

To be seen or to vanish: In the digital economy, visibility is the prerequisite for survival. For decades, that visibility was mediated by ranked lists: firms competed for position on a scrollable page where being first mattered, but being fifth still mattered too. Generative AI search is rewriting this rulebook and it is already reshaping commerce. Adobe Analytics reports that traffic from generative-AI sources to U.S. retail sites rose 1,300% year-over-year in Nov.–Dec. 2024 (Pandya, 2025). Increasingly, consumers receive a synthesized “oracle” response that elevates a select few sources while effectively rendering the vast majority of alternatives invisible. This shift transforms digital competition from how high you rank to whether you are included at all in a brutal, winner-takes-all tournament.

The stakes are immediate for firms. Firms are already reallocating resources from Traditional Search engine optimization (SEO) to generative engine optimization (GEO): investments in machine-readable data, and verifiable signals for ingestion by Large Language Models (Peck, 2025; International Data Corporation, 2025). Yet the rules and returns of this new game remain unsettled. Another industry survey finds 63% of marketers still have no dedicated GEO time, budget, or staff (Goodwin, 2025). In this winner-takes-all tournament, should firms exert surge of effort for the high stake or retreat from a lost cause, risking that GEO may be nothing more than a costly, ineffective hype?

Before writing a guidebook for the players, understanding the market impact is essential. A common concern is a “rich-get-richer” Matthew-effect shift in visibility: Google’s AI Overviews are explicitly constrained to show information “backed up by top web results,” and large-scale audits find that citations skew strongly toward large, high-authority domains (Reid, 2024; Khromova, 2024). While alongside increased concentration among incumbents, there are striking cases where specialized SMEs appear in answers for long-tail, niche queries and capture disproportionate attention and AI-referred visits can be economically meaningful, e.g., 80% higher revenue per visit in travel (Winchell, 2025), leaving real upside for SMEs that clear the inclusion bar (Khromova, 2024)

Consequently, the stakes for platform design are profound, as these algorithms effectively redistribute wealth between industry giants and SMEs. The central design question is: should the search engine prioritize established incumbents to ensure safety, or offer greater opportunities for specialized SMEs to grow?

To address the strategic dilemmas facing firms and the design challenges facing platforms, we develop a model of generative visibility competition. Our framework focuses on how the transition to scarce answer space fundamentally alters the economics of search. By adopting a unified contest-theoretic approach, we analyze the strategic interaction between a high-authority “Head” firm and a low-authority “Tail” firm across both traditional and generative visibility mechanisms.

Our model is built on a simple, empirically grounded distinction: firms are heterogeneous in their prior authority (accumulated reputation and credibility signals) and their intrinsic relevance (true semantic fit to a specific query). To capture the platform’s core design trade-off between “safe” reliance on historical authority and “responsive” adaptability to query-specific

needs, we aggregate these attributes into a single algorithmic score via a single strategic lever: the authority weight. Within this framework, firms exert costly optimization effort to make their intrinsic relevance legible to the algorithm.

Our key results demonstrate a fundamental strategic market bifurcation that explains the coexistence of “rich-get-richer” concentration and long-tail opportunity, and reveal new managerial and policy risks for search platforms and policy makers in the search revolution.

Query-Type Dynamics: In broad queries when firms are similarly relevant to the query (broad, commoditized intent), the algorithm strictly entrenches incumbents behind a wall of legacy reputation and systematically enforces the Matthew Effect. However, in specialized niches, SMEs can override their brand disadvantage through superior fit, capturing the entire market in a single move (long-tail flip). This mechanism provides a precise condition under which specialization becomes a viable escape velocity from legacy authority.

Surge-or-Retreat Logic: In the winner-takes-all tournament, the payoff to effort becomes discontinuous: incremental optimization yields zero return unless the score surmounts the rival’s barrier. Firms therefore face a binary strategic choice: Surge to surpass the rival and win the entire market or Retreat into invisibility to avoid wasted expenditure. This transforms competition from a smooth share-allocation game into a high-stakes tournament driven by entry deterrence, fundamentally altering how resources are deployed.

Incentive Paradox for Firms: Intrinsic relevance and authority are the dual endowments of firms, but they impose opposing forces in firms’ incentives. In Traditional Search, the authority gap acts as a depressant on players’ effort, while Generative Search forces the underdog to exert increasing amounts of effort to deter the head firm’s entry. In Traditional Search, intrinsic relevance acts as a strategic complement to effort, incentivizing the highly-specialized firm to work harder, while in Generative Search, higher relevance acts as a substitute for effort: a more relevant firm can maintain the blockade more easily.

The Safety Paradox for Platforms Our research identifies a profound safety paradox: the authority weight designed to prevent AI hallucinations can inadvertently “freeze” digital markets. While prioritizing established brands seems efficient, it often produces a state of “safe silence” where users are served a reputable name even when a specialized SME offers a superior semantic match. To break this stalemate, platforms must move beyond the zero-sum nature of visibility rewards and incorporate user surplus into their objective

functions. Ultimately, the same “safety knob” used to reduce hallucination risk functions as a powerful market design lever; it dictates market competitiveness, redistributes visibility, and determines the ultimate quality of information delivered to the user.

This paper develops one of the first market-design theories of competition under generative AI search. We model the platform’s scoring rule with a single authority weight that blends legacy authority with query-specific signals, and we nest list-based “soft sharing” and answer-level “hard gatekeeping” within one tractable contest framework. The analysis delivers sharp conditions for when Generative Search entrenches incumbents (the Authority Wall) versus when specialized SMEs can overturn them in niche contexts (the Long-Tail Flip), and why incentives collapse into a binary surge-or-retreat logic. Managerially, the implication is that GEO is not incremental SEO: SMEs win by selecting narrow intents where relevance can clear the gate and by committing enough to secure and defend inclusion. For platforms and policymakers, we establish that the authority weight as a potent market-design lever that jointly determines concentration, entry, and answer quality. By moving beyond “safety-centric” tuning, this research provides a strategic roadmap for dynamic, query-sensitive governance to navigate the high-stakes transition from lists to oracles.

2. Literature Review

This research bridges three streams. First, we extend rank-based visibility and SEO to generative AI search, offering a unified framework for competition and market design under answer-level gatekeeping. Second, we contribute to strategic competition by showing how the shift from divisible market-share game to winner-take-all tournaments reshapes incentives and market structure. Third, we connect to matching and platform design by interpreting the authority weight as a market-design lever that governs entry, concentration, and user welfare.

Rank-Based Search, Algorithmic Ranking, and Competition for Visibility

Search rankings are the mechanism by which digital intermediaries allocate attention. Information-retrieval and learning-to-rank research model ranking as mapping heterogeneous signals into an ordered list (Luce, 1959; Liu, 2009); complementary work shows that interface design shapes demand concentration (long tails vs. superstars) and that relevance is inherently query-contingent (Brynjolfsson et al., 2010; Broder, 2002). We carry this core insight that visibility is economically consequential and interface-dependent into the Generative Search era.

A key substream treats ranking rules as market design, especially in sponsored search and position auctions: rank-contingent allocation and pricing reshape bidding incentives, click allocation, and surplus (Edelman et al., 2007; Varian, 2007; Athey and Ellison, 2011). We extend this insight to the organic/generative setting by studying a different discontinuity—from multi-slot lists that share attention to answer-based interfaces that concentrate it on a single slot.

A second substream studies search engine optimization (SEO) as strategic investment in algorithmic discoverability and its interaction with paid search (Berman and Katona, 2013). We extend this logic to GEO, where answer-level gatekeeping makes returns discontinuous: incremental improvements have little value unless they secure inclusion.

A third line of work models competition for rank as a contest with soft success functions, e.g., Tullock or logit shares, so even lower-ranked firms retain some attention (Tullock, 2001; Luce, 1959). This fits ranked lists; our Traditional Search follows this logic, whereas generative answers impose a hard threshold with discontinuous returns to effort.

Market Competition and Competitive Investment: From Bertrand/Cournot to Tournaments

The second stream is classic market-competition theory, i.e., Cournot and Bertrand and its OM extensions with costly strategic levers that shape demand (Cournot, 1838; Bertrand, 1883). We bring this lens to AI search by replacing price/quantity with visibility: firms invest in optimization effort to raise algorithmic scores and thereby capture demand.

Generative Search answers induce Bertrand-style limit behavior: the winner invests just enough to make the rival’s participation unprofitable. Ranked lists instead allocate attention smoothly, sustaining interior efforts and coexistence. By treating the interface (list vs. answer) as the attention-allocation technology, we show how this design shift alone reshapes competitive intensity, entry incentives, and concentration even with primitives held fixed.

Our setting also connects to contests and tournaments in which agents expend costly effort to win a prize (Lazear and Rosen, 1981; Tullock, 2001). Generative Search makes visibility effectively winner-take-all, but unlike standard tournaments the “score” is shaped jointly by endogenous effort and exogenous endowments (authority and intrinsic relevance), mediated by a platform-chosen weight.

Matching Markets, Search Frictions, and Platform Governance

A third stream views platforms as matching institutions that reduce search frictions yet shape incentives: classic search models emphasize how information affects matching and price dispersion (Stigler, 1961; Burdett and Judd, 1983), matching and market-design work studies how assignment rules govern welfare (Gale and Shapley, 1962; Roth, 2002), and two-sided platform economics highlights that intermediary policies determine who matches with whom and with what incentives (Rochet and Tirole, 2003; Armstrong, 2006; Rysman, 2009).

We contribute to this stream by modeling Generative Search as a matching mechanism with two distinct features (i) intrinsic relevance that is query-contingent and exogenous: the “best match” depends on user intent and semantic fit, and (ii) costly optimization as a signal that makes relevance legible to the platform (Spence, 1973). In AI search, the authority–relevance trade-off is especially salient because hallucination risk and trust concerns push platforms toward authority-heavy policies; alignment work in modern LLM pipelines reflects these safety pressures (Ouyang et al., 2022). The authority weight becomes a market-design lever that sets the inclusion barrier, reshaping entry, concentration, and user welfare.

Taken together, these literatures underscore that interface design is market design. Prior work largely studies ranked lists, where attention is divisible and optimization has smooth returns; generative answers impose gatekeeping, making incentives and market outcomes discontinuous. Our unified model links ranking to competition and market design, providing a tractable foundation for studying how AI search governance reshapes digital market structure.

3. Model

We develop a contest-theoretic model to analyze the equilibrium visibility of firms under two distinct search regimes: Traditional Search (List-based) and Generative Search (Answer-based). We focus on the strategic interaction between firms maximizing visibility and a platform maximizing answer utility under constraints.

3.1. Model Primitives

Consider a digital market with two representative firms $i \in \{H, T\}$ – H for Head Firms (established incumbents) and T for Tail Firms (SMEs), competing for user attention in a specific market category. We focus on this binary structure to reflect the empirical “barbell” distribution of digital markets (Brynjolfsson et al., 2010; Porter, 1980). Firms are heterogeneous in two dimensions:

- Authority Endowment: Firms are classified into two categories based on Authority (A), i.e., A publicly observable signal of firm quality (e.g., brand reputation, backlink profile):
 - Head Firms (H): Firms with high authority endowment (A_H), representing established brands with strong evidential signals such as high-profile citations and media presence (e.g., Wikipedia, Amazon, NYT).
 - Tail Firms (T): specialized Small and Medium Enterprises (SMEs) with low authority endowment (A_T), where $A_H > A_T$.
- Intrinsic Relevance ($R_{iq} \geq 0$): The true semantic alignment between firm i and a specific user query q .

Query Relevance. We differentiate user queries based on the distribution of Intrinsic Relevance (R) across the firm population. Grounded in the established taxonomy of search intent (Broder, 2002), we classify queries into two archetypes:

- Broad Queries (q_B): Relevance is commoditize. These are high-frequency, short-tail queries where the user’s intent is satisfied by a wide range of standard solutions (e.g., “Best project management software”). We model this as $R_H = R_T$.

In this context, both the Head firm (e.g., a Fortune 500 incumbent) and the SMEs (e.g., a generic startup) technically satisfy the query intent. The semantic distance between the query vector and the firm’s product vector is roughly equal for both types. For example, if a user asks for “accounting tools,” both QuickBooks (Head) and a small SaaS tool (Tail) are semantically valid answers.

- Niche Queries (q_N): Relevance is specialize. These are lower-frequency, long-tail queries containing specific constraints such as industry verticals, geography, or unique feature sets (e.g., “Project management tool for agile knitting co-ops”). We model this as $R_T > R_H$.

Tail firms often survive by targeting narrow market segments that Head firms ignore due to scale constraints. For q_N , the Tail firm’s content provides a high-fidelity semantic match (“perfect fit”), whereas the Head firm offers only a generic match (“partial fit”). The LLM recognizes the Tail firm as the specific solution to the specific problem, creating a massive relevance gap. For example: In the query “CRM for gluten-free bakeries,” a generalist giant like Salesforce (Head) has low semantic proximity. However, a niche SME specialized in bakery inventory (Tail) has high semantic proximity.

Optimization Effort. Firms can exert costly effort $e_i \geq 0$ to reveal their relevance to the search algorithm. Effort transforms a firm’s Intrinsic Relevance (R_{iq}) into an observable signal (S_i):

$$S_i(e_i) = R_{iq} \cdot e_i \quad (1)$$

We assume a convex cost function $C(e_i) = \frac{1}{2}ce_i^2$, reflecting the increasing marginal costs of content structuring and technical optimization. function ($C(e) \sim e^2$) by observing the stark difference in complexity between superficial and structural optimization: Low-effort optimization involves trivial formatting, such as implementing basic Schema.org tags or restructuring text into “Question-Answer” pairs. Conversely, high-effort strategies require deep structural investments, such as architecting proprietary databases into complex JSON-LD formats—and incurring the significant costs of securing citations in elite training corpora (e.g., *The Wall Street Journal*). This transition from superficial formatting to verifiable authority drives the non-linear cost structure.

While we acknowledge that optimization technologies may differ in complexity (e.g., structured data for Generative Search vs. keywords for Traditional Search), we assume a constant cost parameter c across regimes. This allows us to isolate the structural impact of the visibility allocation rule itself, unconfounded by exogenous cost differentials ¹.

3.2. The Platform’s Problem: Microfoundations of the Scoring Function

To determine visibility allocation, search engines synthesize heterogeneous signals into a computable score (Liu, 2009; Ouyang et al., 2022). To justify our specific scoring function, we model the search engine as a risk-averse decision-maker. The platform cannot directly observe a firm’s true fit; it must infer it by balancing historical reputation against new information.

The platform calculates a firm’s Algorithmic Score (V_i) by combining two distinct information sources: (i) *The Prior* (A_i): The platform’s existing belief about the firm based on its long-term reputation, domain authority, and citation history. (ii) *The Signal* (S_i): The specific, observable signal generated by the firm for the current query to demonstrate its

¹We assume the cost parameter c remains constant across regimes to isolate the strategic impact of the algorithmic shift (the change in λ and attention allocation). In practice, GEO may be more costly than SEO due to higher technical complexity. However, assuming higher costs for GEO would only amplify our results regarding reduced firm entry and effort, making our baseline assumption of equal costs a conservative benchmark.

relevance. The platform’s final Algorithmic Score V_i is a weighted average of the prior and the signal:

$$V_i(e_i) = \lambda A_i + (1 - \lambda) R_{iq} \cdot e_i \quad (2)$$

where $\lambda \in (0, 1)$ is the authority weight. This parameter serves as the platform’s primary strategic lever to regulate the trade-off between credibility (relying on proven history A_i) and responsiveness (relying on new signals S_i).

We assume the authority weight λ is structurally higher in Generative Search than in Traditional Search ($\lambda_{Gen} > \lambda_{Trad}$). This reflects the distinct economic and technical constraints of Large Language Models (LLMs): In Traditional Search, a “bad link” costs little because users browse multiple options and filter information themselves. In Generative Search, the engine gives a single answer. If that answer is wrong, it is a “hallucination,” causing significant reputational damage. To ensure accuracy, generative models are designed to prioritize information from highly trusted sources (A_i) over new information (S_i).

3.3. Visibility Allocation

The fundamental difference between the two search paradigms lies in how they allocate user attention. In Traditional Search, attention is divisible: users scroll through a list of links (a list of Google results), meaning firms ranked second or third still capture significant traffic. In contrast, Generative Search treats attention as exclusive: the engine synthesizes a single, direct answer (a paragraph of ChatGPT answer), effectively silencing competitors who do not secure the top spot. We model visibility not as a guaranteed reward for quality, but as the outcome of this contest for finite user attention.

Traditional Search (The Soft Contest) In the Traditional Search, the interface returns a scrollable list of ranked links, providing users with $K \geq 10$ clickable options. Empirical literature suggests that while click-through rates decay with rank, they do not drop to zero immediately; users frequently consider multiple options before making a choice. Consequently, we model expected visibility π_i^{Trad} using a standard Softmax (or Tullock) contest function (Luce, 1959; Tullock, 2001). This formulation allows for “partial visibility” where firms share the market in proportion to their scores:

$$\pi_i^{Trad}(e_i) = \frac{\exp(V_i/\mu)}{\sum_{j \in \mathcal{J}} \exp(V_j/\mu)} \quad (3)$$

where $\mathcal{J}$ denotes the set of candidate firms and $\mu > 0$ is a smoothing parameter representing user heterogeneity in browsing depth. A key feature of this contest function is that visibility is strictly positive ($\pi_i^{Trad} > 0$) for any finite score. Thus, even if a firm's score V_i is lower than the market leader's, $\pi_i^{Trad} > 0$. An SME ranking 5th or 6th on a search results page still captures a portion of the demand curve.

Generative Search (The Hard Threshold) In the generative regime, the interface collapses a long list of links into a synthesized answer accompanied by a very short set of cited sources. Let k denote the number of sources explicitly surfaced in the answer (typically in a market with N firms, $k \ll N$). We model this inclusion scarcity as a hard-threshold allocation rule:

$$\pi_i^{Gen}(e_i) = \begin{cases} 1 & \text{if Rank}(V_i) \leq k \\ 0 & \text{if Rank}(V_i) > k \end{cases} \quad (4)$$

The competitive margin therefore shifts from *how high you rank* to *whether you are included*: firms outside the top- k receive negligible economically meaningful attention.

4. Equilibrium Analysis

The transition to Generative Search transforms digital competition from a divisible “market-share game” into a discrete “tournament.” This structural shift drives a paradox of incentives: while the “Winner-Takes-All” interface creates a monopoly prize that should incentivize fierce effort, the algorithm’s prioritization of authority (λ) simultaneously erects a barrier that may stifle it. In this section, we analyze the strategic interplay between Head and Tail firms to resolve this tension and address three fundamental questions:

1. Market Structure: Does the transition to Generative Search entrench incumbent dominance (the Matthew Effect) or democratize visibility for high-fit SMEs (the Long Tail effect)?
2. Displacement Conditions: Under what parametric conditions does superior Intrinsic Relevance (R) override the platform’s Authority bias (λ), allowing specialized SMEs to displace incumbents?
3. Incentive Mechanisms: Do the dual endowments of Relevance and Authority act as reinforcing complements or opposing forces in driving a firm’s incentives for visibility?

4.1. Game Definition

In the competition for visibility between a Head firm (H) and a Tail firm (T), each firm chooses an optimization effort level $e_i \in [0, \infty)$ to influence their generated signal $S_i(e_i) := R_{iq}e_i$. Firms maximize profit, defined as the value of the visibility captured ($\alpha \cdot \pi_i$) minus the cost of effort:

$$\Pi_i(e_i, e_{-i}) = \alpha \cdot \pi_i(e_i, e_{-i}) - \frac{1}{2}ce_i^2 \quad (5)$$

where α represents the total economic value of the query traffic (the prize) and $\pi_i \in [0, 1]$ is the visibility share determined by the search algorithm.

DEFINITION 1 (NASH EQUILIBRIUM). The Nash equilibrium is a set of effort levels, (e_H^*, e_T^*) , such that neither firm can increase its profit by unilaterally changing its effort, given the effort chosen by the competitor. Mathematically, the pair (e_H^*, e_T^*) constitutes an equilibrium if e_i^* maximizes the firm's profit function:

$$e_i^* = \arg \max_{e_i \geq 0} \left(\alpha \cdot \pi_i(e_i, e_{-i}^*) - \frac{1}{2}ce_i^2 \right), \text{ for } i \in \{H, T\} \quad (6)$$

4.2. Traditional Search: the Coexistence Equilibrium

In the Traditional Search, visibility is allocated via a soft contest rule: the expected market share of firm i is given by (3), i.e., $\pi_i^{Trad}(e_i, e_{-i}) = \frac{\exp(V_i/\mu)}{\exp(V_i/\mu) + \exp(V_{-i}/\mu)}$, where $V_i = \lambda A_i + (1 - \lambda)R_i e_i$ per the score function in (2).

LEMMA 1 (Equilibrium Optimization Effort). *In Traditional Search, the optimal effort level e_i^* for firm i is characterized by the unique interior solution to the following implicit fixed-point equation:*

$$\frac{\alpha(1-\lambda)R_i}{\mu} \cdot \pi_i(e_i^*, e_{-i}^*)[1 - \pi_i(e_i^*, e_{-i}^*)] = ce_i^* \quad (7)$$

The LHS of Equation (7) represents the marginal benefit of exerting optimization effort. It is composed of three distinct drivers: (i) α the economic value of gaining visibility; (ii) $\frac{(1-\lambda)R_i}{\mu}$ the efficacy of effort; (iii) $\pi_i(1 - \pi_i)$ the intensity of the contest (incentives are highest when the competition is “tight” with equal market share). The RHS is the marginal effect cost.

As the marginal benefit of optimization is strictly positive on LHS of (7), the optimal strategy must be an interior solution $e_i^* > 0$, which leads to our coexistence equilibrium result:

PROPOSITION 1 (The Coexistence Equilibrium). *In Traditional Search), there exists a unique Nash Equilibrium (e_H^*, e_T^*) where both firms exert strictly positive effort ($e_H^* > 0, e_T^* > 0$) and obtain strictly positive visibility ($\pi_T^* > 0$).*

The model guarantees that firms will always engage in optimization rather than opting out. This mathematical property drives a crucial structural result: in Traditional Search, even a firm with an authority disadvantage ($A_T < A_H$) finds it profitable to compete. This sustains the “Long Tail” of SME visibility observed in the traditional SEO.

However, as we often observe the SMEs visibility is low (low ranking and low visibility) in traditional search results. The following Corollary formalize it and show the primary driver of low SME visibility in Traditional Search is not a lack of effort, but their authority disadvantage.

COROLLARY 1. *In the Coexistence Equilibrium, when the Head firm holds the market lead ($\pi_H > 0.5$), an increase in the Authority Gap ($\Delta A = A_H - A_T$) strictly reduces the Tail firm’s visibility:*

$$\frac{\partial \pi_T^*}{\partial \Delta A} < 0, \quad (8)$$

Furthermore, as the gap becomes insurmountable ($\Delta A \rightarrow \infty$), the Tail firm effectively becomes invisible, i.e., $\lim_{\Delta A \rightarrow \infty} \pi_T^ = 0$.*

4.3. Generative Search (The Gatekeeper Tournament)

In Generative Search, the platform interface restricts visibility to a single answer (the “Answer Box”), fundamentally transforming the competition to a Winner-Takes-All contest: the visibility function becomes a sharp step function (4), i.e., A firm receives 100% visibility if its score V_i strictly exceeds the competitor’s, and 0% otherwise:

$$\pi_i^{Gen}(e_i, e_{-i}) = \begin{cases} 1 & \text{if } V_i(e_i) > V_{-i}(e_{-i}) \\ 0 & \text{if } V_i(e_i) \leq V_{-i}(e_{-i}) \end{cases} \quad (9)$$

Although interfaces may cite $k = 2$ or 3 sources, the salient economic feature is that in a market with N firms, $k \ll N$. The relevant strategic problem for any given firm is competition for inclusion at the margin (e.g., the k -th slot). Our two-firm model captures this local contest between a candidate firm and the marginal incumbent occupying the cutoff position.

LEMMA 2. *In Generative Search, the Tail firm dominates visibility if and only if the net relevance gap generated by optimization, exceeds the algorithmic authority barrier:*

$$R_T e_T - R_H e_H > \frac{\lambda}{1-\lambda} \Delta A \quad (10)$$

The term $\frac{\lambda}{1-\lambda} \Delta A$ represents the “Authority Wall.” The ratio $\frac{\lambda}{1-\lambda}$ acts as a leverage multiplier: as the authority weight $\lambda \rightarrow 1$, this multiplier grows asymptotically, inflating a finite historical gap ΔA into an infinite barrier. The LHS ($R_T e_T - R_H e_H$) measures the net relevance gap generated by optimization. To surpass this Authority Wall, the Tail must generate a massive enough net relevance surplus ($R_T e_T - R_H e_H$).

Unlike Traditional Search, Lemma 2 and the Winner-Takes-All interface lead to GEO effort yields discontinuous payoff: zero return unless it is sufficient to trigger a rank inversion. Consequently, firms face a binary strategic choice: Surge (exert massive effort to barely surpass the rival) or Retreat (exit entirely). This discontinuity triggers an asymmetric Bertrand competition. Instead of a “race to the bottom” on price, firms engage in a “race to the top” on score. This creates a recursive dynamic of score overbidding, where each player exerts just enough additional effort to achieve $V_{new} = V_{old} + \epsilon$, thereby capturing the entire market.

The escalation continues until the weaker firm hits its participation constraint ($\frac{1}{2} c e^2 > \alpha$). The contest resolves into a “Limit Effort” equilibrium: the stronger firm exerts just enough effort to set a target score V^* that is slightly above the weaker firm’s maximum profitable score at its “break-even” point $e_{max} = \sqrt{2\alpha/c}$. The winner optimize only enough to make the challenger’s entry unprofitable.

Case A: Broad Queries. When firms possess symmetric relevance efficiency ($R_H = R_T = R$). Due to authority advantage ($\Delta A = A_H - A_T > 0$), the Head firm enjoys a structural “head start” to capture the entire market: it can theoretically match any score produced by the Tail firm while exerting strictly less effort.

PROPOSITION 2 (The Authority Wall). *In Broad Queries under Generative Search, the Head firm captures 100% visibility. The Tail firm faces an insurmountable Authority Wall and exits the market; The Head firm exerts the minimum effort required to deter entry:*

$$e_H^* = (e_{max} - \frac{\lambda}{1-\lambda} \frac{\Delta A}{R_H})^+, \quad e_T^* = 0. \quad (11)$$

When firms compete on generic topics where relevance is commoditized ($R_H \approx R_T$), the algorithm strictly enforces the **Matthew Effect**: the Head firm captures the entire market. Because the SMEs lack the relevance surplus necessary to breach the Authority Wall. This result addresses our first research question: For broad queries, rather than democratizing visibility, the generative interface systematically entrenches the pre-existing market structure. Furthermore, Eq. (11) reveals a substitution effect between authority and effort. As the incumbent's authority advantage (ΔA) increases, their equilibrium effort e_H^* strictly decreases.

Case B: Niche Queries Consider niche queries where the Tail firm has a relevance advantage ($R_T > R_H$). Let $\Delta R = R_T - R_H > 0$ represent the “specialization premium.” Different from broad queries, in niche queries, optimization effort e_T is multiplied by the high intrinsic relevance R_T in (10). This creates the “Long Tail” opportunity: by engaging in GEO, the Tail firm can win if its efficiency in generating relevance compensates for its deficit in authority.

THEOREM 1 (Long-Tail Flip). *In Niche Queries, the specialized SME(T) displaces the Incumbent(H) if and only if the “Relevance Surplus” generated at maximal effort (e_{max}) overcomes the Authority Wall:*

$$\underbrace{(R_T - R_H)e_{max}}_{\text{Net Relevance Surplus}} > \underbrace{\frac{\lambda}{1 - \lambda}\Delta A}_{\text{Authority Wall}}, \quad (12)$$

Under this condition, the unique Nash Equilibrium involves the SME capturing 100% visibility and setting an equilibrium effort e_T^ that deters the Head firm's entry ($e_H^* = 0$):*

$$e_T^* = \frac{\lambda}{1 - \lambda} \frac{\Delta A}{R_T} + \frac{R_H}{R_T} e_{max}. \quad (13)$$

Inequality (12) provides the precise parametric answer to the displacement question. It establishes that for Intrinsic Relevance to mathematically override the platform's Authority bias (λ), the SME must exhibit hyper-specialization. Specifically, the relevance gap ($R_T - R_H$) must be sufficiently large to satisfy the Authority Wall constraint. The equilibrium effort e_T^* functions as a strategic deterrent: by raising the performance threshold, the SME forces the Head firm into a zone where the cost of matching visibility exceeds the potential payoff, thereby inducing their exit.

Proposition 2 and Theorem 1 together demonstrate the fundamental asymmetry of the generative algorithm, defining two distinct regimes of competition, and highlights the SMEs opportunity:

- *Broad Queries (Authority Wall)*: Where relevance is commoditized ($R_H \approx R_T$), Authority (ΔA) dictates the winner. The Head firm wins by default, not through superior fit, but through the structural weight of their legacy.
- *Niche Queries (Long-Tail Flip)*: Where relevance is highly differentiated, the SME leverages superior relevance to counterbalance its authority deficit. By generating a sufficient relevance surplus via GEO, the SME vaults the Authority Wall to displace the Incumbent(H). This mathematical bifurcation explains the mechanism behind the empirical observation that SME visibility probability increases significantly in long-tail contexts (ZipTie.dev, 2024; BrightEdge, 2024).

4.4. Structural Transformation: Coexistence vs. Disruption

The transition from Traditional to Generative Search represents a fundamental shift in market structure, replacing a regime of continuous oligopoly with one of discontinuous disruption.

In Traditional Search (Figure 1, Blue Dashed), the market operates as a regime of competitive coexistence. Visibility is elastic: even facing a significant authority gap ($\lambda = 0.65$), a high-effort SME secures a foothold, with market share scaling smoothly against relevance. This mechanism sustains a vibrant “Long Tail,” allowing niche firms with moderate relevance advantages to survive alongside the Head firms.

In Generative Search (Figure 1, Red Solid), this stability is replaced by a binary “tipping point” dynamic. The market exists in one of two states: total blockade or total disruption. A higher authority weight ($\lambda = 0.80$) constructs an Authority Wall that functions as an absolute barrier, rendering the SME invisible regardless of their effort. However, once the SME’s relevance surplus exceeds the barrier, the equilibrium flips instantaneously, and the SME captures the entire market.

Underlying this structural shift is a fundamental tension between the two assets firms possess. We now turn to these dual endowments to examine how intrinsic relevance and accumulated authority drive the opposing incentives in the competition for attention.

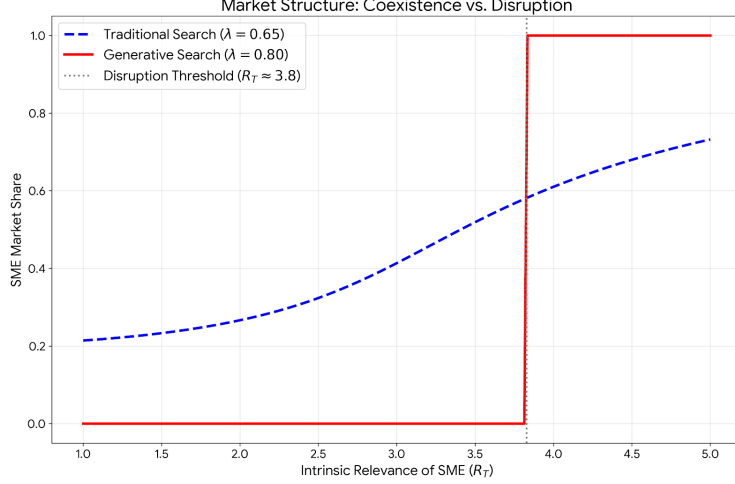


Figure 1 SME Market Share. *SEO: Blue Dashed ; GEO: Red Solid* with Disruption Threshold $R_T \approx 3.8$.

5. Dual Endowments: The Opposing Forces of Relevance and Authority

We now turn to the critical tension driving these market dynamics: the firm's dual endowments. In the scoring equation $V_i(e_i) = \lambda A_i + (1 - \lambda) R_{iq} \cdot e_i$, Intrinsic Relevance (R) and Authority (A) appear to play similar roles as both boost the score and aid visibility. Yet, this algebraic similarity masks a deeper economic conflict. When we analyze the incentives for optimization, we uncover a striking paradox: these two assets exert opposing forces on the decisions. Unraveling this conflict is key to understanding why the transition to Generative Search fundamentally alters the logic of market competition.

5.1. Relevance: Strategic Complement in SEO vs. Substitute in GEO

In Traditional Search, relevance acts as a strategic complement to effort. Because the marginal productivity of optimization is increasing in relevance ($\frac{\partial^2 V}{\partial e \partial R} > 0$), the algorithm naturally incentivizes the “fitter” firm to work harder.

PROPOSITION 3. *In Traditional Search, the ratio of optimization efforts is proportional to the ratio of intrinsic relevance.*

$$\frac{e_T^*}{e_H^*} = \frac{R_T}{R_H} \quad (14)$$

This implies that optimization is allocatively efficient: the firm with superior relevance (R_i) exerts proportionally higher effort. As shown in Figure 2 (Panel A, Blue Dashed Curve), effort rises steeply with relevance.

Generative Search breaks this connection between intrinsic relevance and effort. Once the SME breaches the Authority Wall, the strategic objective shifts from visibility competition to entry deterrence. As derived in Theorem 1, the winner's optimal effort becomes inversely proportional to their relevance:

$$e_T^* = \frac{1}{R_T} \left(\frac{\lambda}{1-\lambda} \Delta A + R_H e_{max} \right)$$

Here higher relevance acts as a substitute for effort: a more relevant firm can maintain the blockade more easily. As shown in Figure 2 (Panel A, Red Solid Curve), this leads to a sharp incentive reversal—once the market is captured, further improvements in relevance allow the winner (SME) to reduce effort while maintaining total dominance.

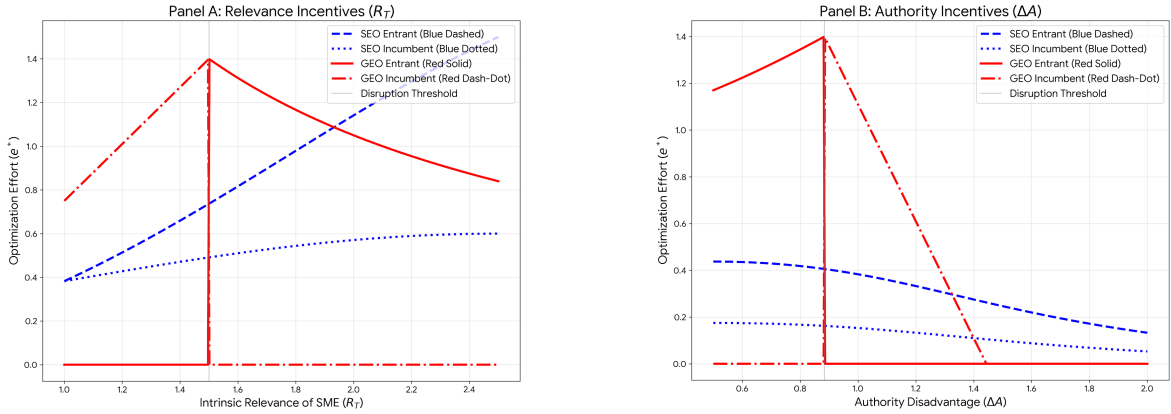


Figure 2 The Dual Endowments. The opposing incentives of Intrinsic Relevance (A) and Authority (B).

5.2. Authority: The Discouragement Effect in SEO vs. The Escalation Trap in GEO

While relevance shifts from a strategic complement in SEO to a substitute in GEO, authority functions as a structural constraint that dictates the threshold of competition. The impact of the Authority Gap ($\Delta A = A_H - A_T$) on firm incentives reveals the most profound divergence between the two search mechanisms.

In Traditional Search, once the Head firm establishes dominance ($\pi_T < 0.5$), authority acts strictly as a depressant on effort as shown in Proposition 4 and Figure 2 (B, Blue Curves).

PROPOSITION 4. *In Traditional Search, when the Head firm holds the market lead ($\pi_H > 0.5$), widening the authority gap strictly reduces SEO effort for both firms:*

$$\frac{\partial e_T^*}{\partial \Delta A} < 0 \quad \text{and} \quad \frac{\partial e_H^*}{\partial \Delta A} < 0 \quad (15)$$

A large authority gap makes the market less competitive by suppressing both firms' SEO effort: the SME (T) rationally pulls back because the marginal return on fighting a “lost cause” is too low. Simultaneously, the Head firm (H) reduces effort because their victory is effectively assured by their brand history (A_H).

Generative Search fundamentally alters this dynamic. In the binary tournament, when the SME wins the market, the authority gap functions as an entry threat from the high-authority Head firm. As ΔA widens, the specialized SME is forced to escalate the competition, setting an increasingly aggressive performance threshold to deter the Head firm's entry as illustrated in Figure 2 (B, Red Solid). The SME's effort scales linearly with the authority gap (ΔA) is derived in Theorem 1:

$$e_T^* = \frac{\lambda}{1 - \lambda} \frac{\Delta A}{R_T} + \frac{R_H}{R_T} e_{max} \quad (16)$$

Unlike in Traditional Search, where a large disadvantage leads to resignation, Generative Search forces the underdog to exert increasing amounts of effort just to remain in the game.

5.3. Opportunities and Risks for SMEs in the Generative Era

In such a fatal binary tournament for visibility, challenges also come with opportunities for SMEs. While niche queries may enable a “long-tail flip,” market potential remains heterogeneous across different verticals such as scale and profitability. We now identify the specific high-value niches that provide real opportunities for specialized SMEs.

High-Value Niche When the value of visibility α increases, firms are incentivized to increase their effort. In Traditional Search, the growing of the economic stake leads to a continuous increase of optimization effort. In Generative Search, rather than gradual adaptation, SMEs remain static silent ($e_T^* = 0$) and invisible until the prize α hits a critical threshold ($\bar{\alpha}$). It triggers a sudden “long-tail flip” and a massive surge in effort ($e_T^* \geq \sqrt{\frac{2\bar{\alpha}}{c}}$). This creates abrupt optimization shocks as firms jump from total inaction to hyper-aggressive competition once the niche becomes profitable enough to justify the entry cost.

COROLLARY 2 (Critical Prize Threshold). *In Generative Search, there exist a critical prize threshold $\bar{\alpha}$, such that the Tail firm’s equilibrium effort level follows*

$$e_T^* = \begin{cases} 0 & \text{if } \alpha < \bar{\alpha} \text{ (Retreat)} \\ \frac{R_H}{R_T} \sqrt{\frac{2\bar{\alpha}}{c}} + \frac{\lambda}{1-\lambda} \frac{\Delta A}{R_T} & \text{if } \alpha \geq \bar{\alpha} \text{ (Surge)} \end{cases} \quad (17)$$

where $\bar{\alpha} = \frac{1}{2} \left[\frac{\lambda \Delta A}{(1-\lambda)(R_T - R_H)} \right]^2 c$, and consequently, $e_T^* \geq \sqrt{\frac{2\bar{\alpha}}{c}}$.

SMEs Opportunities and Risks:

These high-value niches represent the primary frontier for specialized SMEs who historically struggled with low visibility and declining market share in Traditional Search. In the generative era, these firms have a unique opportunity to dominate their vertical markets by leveraging superior fit to achieve total visibility. However, this opportunity is governed by three critical strategic constraints:

1. **Surge or Retreat:** Success in a high-value niche demands absolute commitment rather than incremental improvement. Firms cannot “gradually” improve their way into the answer box; they must exert enough effort to surmount the Authority Wall immediately, or they remain effectively invisible to the user.
2. **Continuous Deterrence:** Entry into the answer slot is only the first step. Once an SME achieves the “long-tail flip,” it must continuously exert enough effort to safeguard its position, because competition for the single answer slot is essentially a fatal tournament.
3. **Platform Diversification:** SMEs should pivot resources toward emerging AI engines and vertical marketplaces. These platforms typically utilize a lower authority weight (λ), favoring specific query relevance over legacy domain history. By targeting these more meritocratic environments, specialized firms can bypass the Authority Wall of dominant incumbents and achieve a sustainable return on their optimization effort (GEO).

6. Managerial and Policy Implications for the Platform

We now turn to the search platform’s primary strategic lever: the authority weight (λ). While common wisdom suggests that platforms tune this weight solely as a safety mechanism to reduce hallucinations, our model reveals it effectively functions as a wealth redistribution lever between big giants and SMEs. In the generative era, it dictates which firms survive and which are silenced. Here we demonstrate that designing this weight requires moving beyond firm-level competition to prioritize search engines’ ultimate stakeholder: the user, and why.

6.1. Distributional Impact: The “Rich Get Richer”

We first examine how the platform’s authority weight λ impacts the profit of Head and Tail firms.

A. Traditional Search. In Traditional Search, as the platform places more weight on historical authority (A) and less on effort (e), the marginal productivity of optimization declines. This creates a natural surplus for the Head firm.

PROPOSITION 5. *In Traditional Search, let Π_H^* and Π_T^* denote the equilibrium profits of the Head and Tail firms, respectively. An increase in the authority weight (λ) strictly increases the absolute profit gap between the firms, given λ is sufficiently large:*

$$\frac{\partial(\Pi_H^* - \Pi_T^*)}{\partial\lambda} > 0. \quad (18)$$

This proposition establishes that increasing the authority weight (λ) strictly widens the profit gap in favor of the Head firm. This occurs because a higher λ amplifies the value of historical endowment which is the Head firm’s comparative advantage, while simultaneously eroding the efficacy of the entrant’s optimization effort.

B. Generative Search. The binary allocation of visibility transforms the Authority Wall ($\frac{\lambda}{1-\lambda}\Delta A$) into a strict gatekeeper that asymmetrically impacts profit incentives.

- **Broad Queries:** For the incumbent(H), a higher λ fortifies the Authority Wall, effectively acting as an algorithmic subsidy. This allows the Head firm to capture the full prize α with minimal defensive effort,
- **Niche Queries:** Even with superior relevance ($R_T \gg R_H$), the SME’s profit is strictly decreasing in λ . To maintain the “long-tail flip,” the SME is forced to offset the Head firm’s authority advantage:

$$\Pi_T^* = \alpha - \frac{1}{2}c \left(\underbrace{\frac{\lambda}{1-\lambda} \frac{\Delta A}{R_T}}_{\text{Authority Wall}} + \frac{R_H}{R_T} e_{max} \right)^2 \quad (19)$$

As λ increases, the SME must exert strictly higher effort to deter the Head firm’s entry. This mandatory expenditure on the Authority Wall directly erodes profit (Π_T^*).

These results fundamentally reframe algorithmic governance: the authority weight (λ) functions less as a technical safeguard against errors and more as an instrument of wealth redistribution. By systematically widening the profit gap, a high λ transfers surplus from high-relevance entrants(T) to high-authority incumbents(H). Consequently, determining λ is not merely an engineering decision, but a policy choice that prioritizes the protection of historical capital over the reward of current merit.

6.2. The Planner’s Dilemma: The Efficiency Trap

From the perspective of total industry surplus, the optimal strategy seems simple: set the authority weight to the maximum ($\lambda \rightarrow 1$). This stems from the zero-sum nature of the visibility prize (α). Since total industry revenue is fixed, maximizing industry surplus is equivalent to minimizing total effort. Setting $\lambda = 1$ achieves this “perfect efficiency” by rendering the Authority Wall insurmountable: the Incumbent wins by default without effort ($e_H^* = 0$), and the SMEs rationalizes exit ($e_T^* = 0$), driving total effort cost to zero.

While efficient for the market, this creates a market of “safe silence.” The incumbent and the SME have no incentive to exert effort to reveal relevance. Users are served the “safe” brand even when a specialized SME offers a perfect semantic match (R_T), degrading the utility of the search engine itself.

Maximizing total firms’ profit leads to a perverse outcome where no firm exerts effort to reveal relevance, ignoring the ultimate source of value: the users’ need for an relevant answer. To avoid this trap, our objective function must incorporate user utility into the objective.

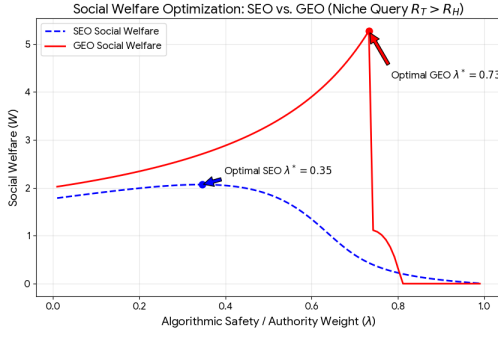
6.3. Social Welfare Optimization

We propose a Social Welfare Function (W) that balances the societal value of high-relevant answers against the economic cost of producing them.

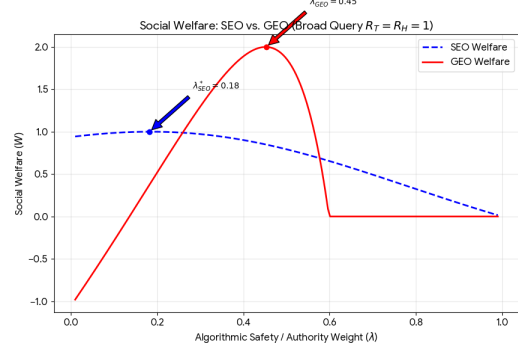
$$\max_{\lambda \in [0,1]} W(\lambda) = \beta \cdot \mathbb{E}[R_{winner} \cdot e_{winner}^*] - \frac{1}{2} \sum_{i \in \{H,T\}} c(e_i^*)^2 \quad (20)$$

Where β represents user utility weight on the relevance of the winner; $R_{winner} \cdot e_{winner}^*$ is the realized relevance of the visible answer; and $\frac{1}{2}c(e^*)^2$ is the total loss of optimization effort. Note that the prize (α) captured by the winning firm is excluded from this welfare formulation, as it represents a constant transfer payment that does not vary with λ .

We analyze the welfare implications of the transition from Tranditional Search to Generative Search across two distinct market regimes: Niche Queries ($R_T > R_H$) and Broad Queries ($R_T \approx R_H$).



(a) Niche Queries ($R_T > R_H$)



(b) Broad Queries ($R_T \approx R_H$)

Figure 3 Social Welfare Analysis.

Niche Queries (Figure 3(a)). In markets dominated by specialized intent ($R_T = 3.0, R_H = 1.0$), the result reveals a “high risk, high reward” dynamic for GEO:

- In Traditional Search, the welfare curve (Blue Dashed) is stable but strictly lower than the GEO peak. The optimal authority weight is low ($\lambda_{SEO}^* \approx 0.35$), as the platform prioritizes a balanced ecosystem where the high-relevance SME retains enough visibility to justify moderate effort. However, the fragmented nature of the ranked list dilutes the “winner” prize, preventing the SME from exerting maximum optimization effort.
- Generative Search (Red Solid) achieves higher welfare and peaks at $\lambda_{GEO}^* \approx 0.73$. Raising the Authority Wall with increasing λ compels the specialized SME to maximize effort to maintain visibility. This mechanism effectively extracts maximum quality (e^*) from the entrant, benefiting users. However, exceeding this threshold ($\lambda > 0.73$) forces the SME to exit. The market flips to the Head firm ($R_H \ll R_T$), causing welfare to plummet from $W \approx 5.2$ to $W \approx 1.2$. Once the threat of entry is eliminated ($\lambda \approx 0.81$), the incumbent(H)’s deterrence incentives evaporate. As their effort e_H^* drops to zero (Proposition 2), social welfare fades entirely to zero.

Broad Queries (Figure 3(b)). In markets with commoditized relevance ($R_T \approx R_H$), Generative Search offers a higher potential welfare ceiling but requires precise algorithmic tuning to avoid market failure.

- The GEO curve (Red Solid) peaks significantly higher ($W \approx 2.0$) than SEO ($W \approx 1.0$). While Traditional Search creates a “safe” equilibrium where fragmented attention limits

optimization, Generative Search’s “Winner-Take-All” visibility can incentivize the Head firm to exert the socially optimal level of effort, provided the platform sets the correct resistance ($\lambda_{GEO}^* \approx 0.45$).

- Unlike SEO, Generative Search is volatile. While GEO offers higher potential welfare, it requires precise tuning ($\lambda_{GEO}^* \approx 0.45$) to manage the Head firm’s incentives who wins in broad queries, λ determines their effort level to deter entry: If λ is too low, the negligible Authority Wall forces the incumbent into excessive deterrence effort which significantly lowers social welfare. If λ is too high, the insurmountable Authority Wall allows the incumbent to secure the market with little effort thus hurt users’ surplus.

6.4. Policy Recommendation: Dynamic Governance

Our analysis offers three critical insights for platform governance. First, while Generative Search distributes visibility less evenly than ranked lists, its “Winner-Takes-All” mechanism concentrates incentives, potentially achieving superior social welfare. Second, this high-efficiency outcome is fragile; slight miscalibration can trigger market collapse. Third, the authority weight (λ) must therefore be treated not merely as a safety parameter against hallucinations, but as a strategic lever to ensure user utility. We propose a dynamic governance approach:

- (a) **Niche Queries:** Favor generative answers, but choose a *relatively high yet subcritical* λ that still permits specialized entry. Increasing λ can discipline inclusion and raise the realized relevance of the winning specialist (via higher equilibrium effort) However, governance must be precise: exceeding the critical threshold eliminates the entrant, causing welfare to vanish.
- (b) **Broad Queries:** Choose based on Tuning Capability. A well tuned Generative Engine can outperform ranked lists. By calibrating the Authority Wall, the platform can prevent the market from overheating into wasteful rivalry, while maintaining the competitive tension necessary to drive quality. When such tuning is infeasible or unstable, ranked lists provide a robust baseline by preserving partial visibility and competitive pressure under imperfect parameterization. If precise tuning is infeasible, ranked lists offer a robust safety net. By sharing the spotlight among multiple rivals, Traditional Search keeps competition active by default, preventing the market from freezing into a stagnant monopoly, even without perfect algorithmic calibration.

Implementation feasibility. Dynamic governance is operationally feasible because it extends existing “universal search” pipelines: modern engines already classify query intent in real time and route queries to different modules (e.g., AI answers vs. blue links). The same architecture can condition λ (and k) on predicted “broad” vs. “niche” intent, enabling query-sensitive tuning without a redesign of the entire ranking stack.

7. Conclusion

The transition from list-based search to generative AI “oracles” represents a structural shift in digital visibility: traditionally, ranked lists allowed firms to coexist across multiple positions, but Generative Search compresses results into single, synthesized answers—firms are either mentioned or omitted. We formalize this evolution through a contest-theoretic model to analyze the strategic interaction between firms and platforms under these two distinct visibility mechanisms.

At the heart of this transition is the conflict between incumbent stability and meritocratic disruption. Our analysis addresses whether the generative “winner-takes-all” tournament inevitably facilitates a “rich-get-richer” Matthew Effect or empowers agile SMEs to flip the market. We find that the architecture of the search interface dictates the nature of competition: while Traditional Search permits a coexistence Equilibrium where firms with authority disadvantages maintain marginal visibility, Generative Search imposes a “hard threshold” that forces firms into a binary choice of aggressive Surge or total Retreat.

Generative search transforms market dynamics from stable coexistence into a winner-take-all tournament dictated by query context. In Broad Queries ($R_T \approx R_H$), an insurmountable Authority Wall systematically entrenches incumbent dominance, allowing the Head firm to capture the market by default. Conversely, in Niche Queries ($R_T > R_H$), LLM semantic sensitivity creates a “relevance escape.” By leveraging hyper-specialization, a Tail firm can achieve a long-tail flip, overriding its authority deficit to seize the market and displace the incumbent.

For management practitioners, particularly in the SME sector, the generative era requires a shift toward deep specialization and “all-in” commitment. Because AI search interface imposes a binary winner-take-all tournament, marginal efforts result in total invisibility; firms must target narrow niches where their unique relevance far exceeds a big brand’s generic

advantage. Once they capture the single answer slot, specialists must maintain a high enough score to deter entry.

Finally, our research highlights critical implications for search platforms and policymakers regarding the authority weight. While often viewed purely as a safety mechanism to reduce hallucinations, it is a powerful tool that shifts wealth between industry giants and small players. To maximize social welfare, platforms must move beyond safety-centric tuning to incorporate user utility, balancing the societal value of highly relevant answers against the economic cost of optimization. We propose a dynamic governance strategy to maximize social welfare: in specialized niches, set a high authority weight to incentivize firms to provide their best possible answer for users. For broad contexts, use Generative Search for maximum efficiency if it is perfectly tuned, otherwise relying on traditional lists as a safety net to prevent monopoly stagnation. This shift toward welfare-centric tuning ensures the search ecosystem remains competitive while delivering the highest utility to users.

References

- Armstrong, M. (2006). Competition in two-sided markets. *RAND Journal of Economics* 37(3), 668–691.
- Athey, S. and G. Ellison (2011). Position auctions with consumer search. *Quarterly Journal of Economics* 126(3), 1213–1270.
- Berman, R. and Z. Katona (2013). The role of search engine optimization in search marketing. *Marketing Science* 32(4), 644–651.
- Bertrand, J. (1883). Théorie mathématique de la richesse sociale [book review]. *Journal des Savants* 67, 499–508.
- BrightEdge (2024, September). Google AIO is ranking more niche specific sites. Search Engine Journal. <https://www.searchenginejournal.com/google-aio-is-ranking-more-niche-specific-sites/528710/>. Data showing specialized sites gaining visibility over generalists in AI Overviews.
- Broder, A. Z. (2002). A taxonomy of Web search. *ACM SIGIR Forum* 36(2), 3–10.
- Brynjolfsson, E., Y. J. Hu, and M. D. Smith (2010). Research commentary—long tails vs. superstars: The effect of information technology on product variety and sales concentration patterns. *Information Systems Research* 21(4), 736–747.

- Burdett, K. and K. L. Judd (1983). Equilibrium price dispersion. *Econometrica* 51(4), 955–969.
- Cournot, A. A. (1838). *Recherches sur les principes mathématiques de la théorie des richesses*. Paris: L. Hachette.
- Edelman, B., M. Ostrovsky, and M. Schwarz (2007). Internet advertising and the generalized second-price auction: Selling billions of dollars worth of keywords. *American Economic Review* 97(1), 242–259.
- Gale, D. and L. S. Shapley (1962). College admissions and the stability of marriage. *The American Mathematical Monthly* 69(1), 9–15.
- Goodwin, D. (2025, September). Marketers aren’t ready for GEO: Survey. Search Engine Land. <https://searchengineland.com/marketers-arent-ready-for-geo-survey-461919>. Survey results shared by Centerfield; accessed 2025-12-26.
- International Data Corporation (2025, September). Marketing’s new imperative: The shift from seo to llm optimization. IDC Blog. <https://www.idc.com/resource-center/blog/marketings-new-imperative-the-shift-from-seo-to-llm-optimization/>. Accessed: 2025-12-26.
- Khromova, Y. (2024, December). The state of AI overviews in 2024: Research insights and future forecasts. SE Ranking Blog (Studies & Surveys). <https://seranking.com/blog/ai-overviews-2024-recap-research/>. Large-scale audit of Google AI Overviews regarding trigger rates and domain concentration. Accessed 2025-12-26.
- Lazear, E. P. and S. Rosen (1981). Rank-order tournaments as optimum labor contracts. *Journal of Political Economy* 89(5), 841–864.
- Liu, T.-Y. (2009). Learning to rank for information retrieval. *Foundations and Trends in Information Retrieval* 3(3), 225–331.
- Luce, R. D. (1959). *Individual choice behavior: A theoretical analysis*. New York: John Wiley & Sons.
- Ouyang, L., J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Aspell, P. Welinder, P. Christiano, J. Leike, and R. Lowe (2022). Training language models to follow instructions with human feedback. arXiv preprint arXiv:2203.02155. <https://arxiv.org/abs/2203.02155>. Accessed 2025-12-26.

- Pandya, V. (2025, November). Adobe: Consumers spent \$88.7 billion online (oct 2025), up 8.2% year-over-year, with AI-powered shopping continuing to rise. Adobe for Business Blog (Adobe Analytics / Adobe Digital Insights). <https://business.adobe.com/blog/adi-october-2025-holiday-shopping-actuals>. Reports surge in generative-AI traffic to U.S. retail sites and compares conversion and revenue per session for AI-referred vs. non-AI traffic. Accessed 2025-12-26.
- Peck, A. (2025, November). Search marketing budgets 2025: GEO vs. SEO & PPC. Clutch.co (Clutch Report). <https://clutch.co/resources/geo-vs-seo-ppc-survey-report>. Survey of 600 marketing professionals (fielded Aug 2025); accessed 2025-12-26.
- Porter, M. E. (1980). *Competitive strategy: Techniques for analyzing industries and competitors*. New York: Free Press.
- Reid, E. (2024, May). AI overviews: About last week. The Keyword (Google). <https://blog.google/products/search/ai-overviews-update-may-2024/>. Google product blog post describing how AI Overviews are integrated with web ranking systems. Accessed 2025-12-26.
- Rochet, J.-C. and J. Tirole (2003). Platform competition in two-sided markets. *Journal of the European Economic Association* 1(4), 990–1029.
- Roth, A. E. (2002). The economist as engineer: Game theory, experimentation, and computation as tools for design economics. *Econometrica* 70(4), 1341–1378.
- Rysman, M. (2009). The economics of two-sided markets. *Journal of Economic Perspectives* 23(3), 125–143.
- Spence, M. (1973). Job market signaling. *Quarterly Journal of Economics* 87(3), 355–374.
- Stigler, G. J. (1961). The economics of information. *Journal of Political Economy* 69(3), 213–225.
- Tullock, G. (2001). *Efficient Rent Seeking*, pp. 3–16. Boston, MA: Springer US.
- Varian, H. R. (2007). Position auctions. *International Journal of Industrial Organization* 25(6), 1163–1178.
- Winchell, A. (2025, May). The explosive rise of generative AI referral traffic. Adobe for Business Blog (Adobe Digital Insights). <https://business.adobe.com/blog/the-explosive-rise-of-generative-ai-referral-traffic>. Adobe Digital Insights report on growth and value of AI-driven referral traffic. Accessed 2025-12-26.

ZipTie.dev (2024, October). State of AI overviews: 5 disruptions found after analyzing 500k queries. ZipTie.dev. <https://ziptie.dev/blog/state-of-ai-overviews-5-google-aio-disruptions-found-after-analyzing-500k-queries/>. Report finding 63% of AI Overview sources do not rank in the traditional top 10.